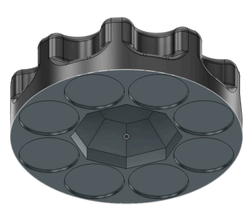
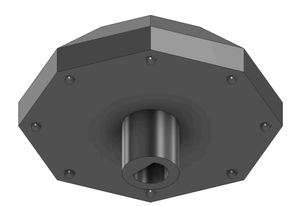
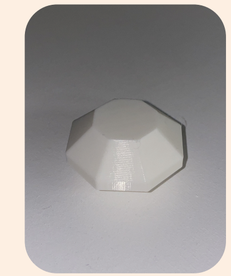
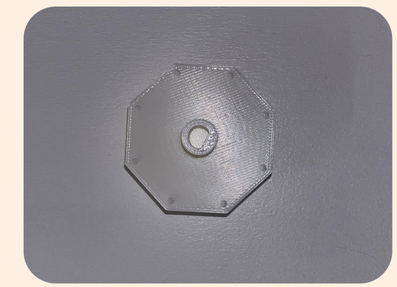
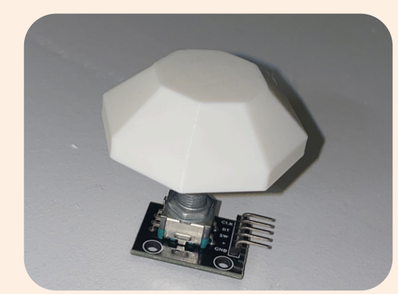
Technical - Design of the Solution

Parts:
- Desktop
- Tangible (Grip with animal on top)
- Tangible Base / Adapter
- Rotery-Encoders
- RFID- Readers (NFC-Readers)
- NFC-Chips
- Arduino ESP32-S3 (Dev Module)

The prototype consists of a custom made Desktop with holes drilled into it. The holes connect the turning-mechanism on top, which also is the base for the tangible, to the eletronics below the desktop.

The tangibles have a octagon shape cut out, with slants at the bottom. The ports are a smaller shape of octagon . Because of this the tangible can slide from all possible directions on the base in order to not restict the user to place the tangibles in a certain way.
The edges on the octagon extrusion catch the edges of the cut-out when the tangible is being turned. This is how the rotation is transfered to the encoder, via a custom shape cut-out at the bottom of the port. The enconders are screwed onto the desk from below and are connected to the arduino.